The empirical recurrence for OEIS A232379, recovered from its tail

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Abstract

OEIS A232379 records "Empirical recurrence of order 60" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 223$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 62$. Its last coefficient is $c_{60} = 6656 \neq 0$, so no further tail satisfies anything shorter; any order-60 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A232379 is "Number of 4Xn 0..3 arrays with every 0 next to a 1, every 1 next to a 2 and every 2 next to a 3 horizontally, diagonally or antidiagonally, and no adjacent values equal.". It has offset 1 and begins

$$1, 230, 4550, 63744, 927706, 14326374, 219597458, 3342549408, \ldots$$

The entry states:

> Empirical recurrence of order 60 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 07:22:43, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 223$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 223$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 60: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 60.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 446$ exact terms from $n = 2$ on, it returns order 60 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{60} c_i\, a(n-i),$$

c_1	14	c_{16}	132823819	c_{31}	−5378345413	c_{46}	−2343274501
c_2	−4	c_{17}	−20714248	c_{32}	−277248001	c_{47}	212504447
c_3	336	c_{18}	−421637456	c_{33}	6550559022	c_{48}	840361375
c_4	235	c_{19}	−1441135408	c_{34}	3988115328	c_{49}	62167723
c_5	509	c_{20}	−2085064311	c_{35}	−9633041858	c_{50}	−161249357
c_6	−6502	c_{21}	−2573603617	c_{36}	−8001048556	c_{51}	−33191904
c_7	−504	c_{22}	−2651217009	c_{37}	8248974297	c_{52}	−21969008
c_8	48298	c_{23}	−1807337624	c_{38}	7976667172	c_{53}	−18214676
c_9	−725960	c_{24}	−3005061193	c_{39}	−2520890818	c_{54}	15784728
c_{10}	−874046	c_{25}	−4368677349	c_{40}	−4952285830	c_{55}	14342480
c_{11}	−1597129	c_{26}	−3876772249	c_{41}	−3987706835	c_{56}	−406496
c_{12}	538619	c_{27}	−1247515383	c_{42}	−1088874919	c_{57}	−2150656
c_{13}	22402203	c_{28}	−299365715	c_{43}	3714413801	c_{58}	−262272
c_{14}	54634448	c_{29}	−1880595889	c_{44}	2883237975	c_{59}	56320
c_{15}	124941593	c_{30}	−5291086231	c_{45}	−1849937340	c_{60}	6656

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 62$, and it is the recurrence OEIS A232379 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{60} - c_1 x^{60-1} - \cdots - c_{60}$. Its constant term is $-c_{60} = -6656$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 60 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 60, so $\deg q = 60$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 60, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 60, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 6656.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-60 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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